TWO CATEGORICAL VARIABLES

new 2.1–2.2 • old 2.1–2.3

ZONE A • COUNTS, SHARES, AND COMPARISONS

DO STUDENTS HAVE A JOB?

	Grade 11	Grade 12	Total
Job	12	36	48
No job	28	24	52
Total	40	60	100

Within Grade 11: $12/40 = 30\%$ have a job.
Within Grade 12: $36/60 = 60\%$ have a job.

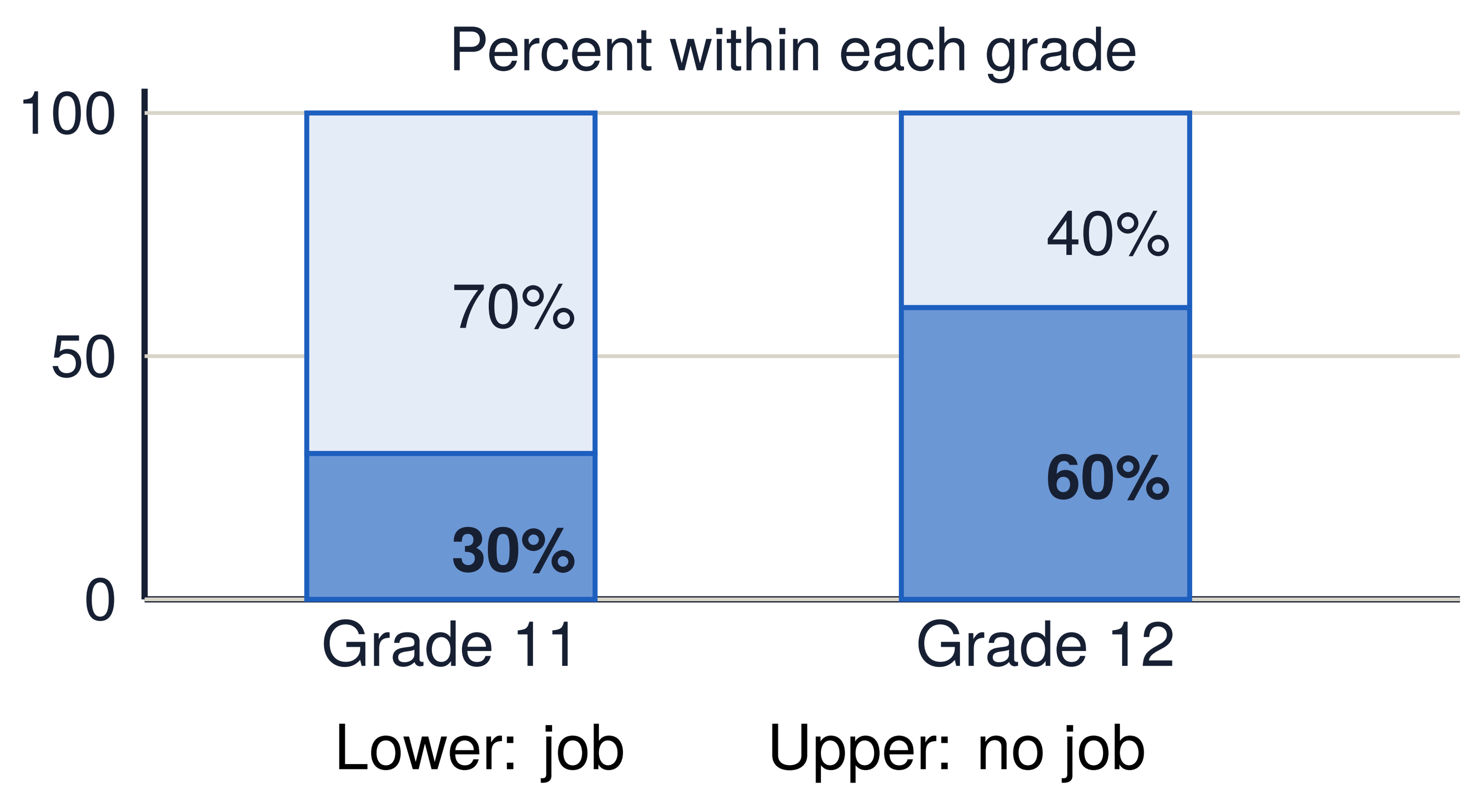
CHOOSE THE DENOMINATOR

$$\text{joint rel. freq.} = \frac{\text{cell}}{\text{grand total}}$$

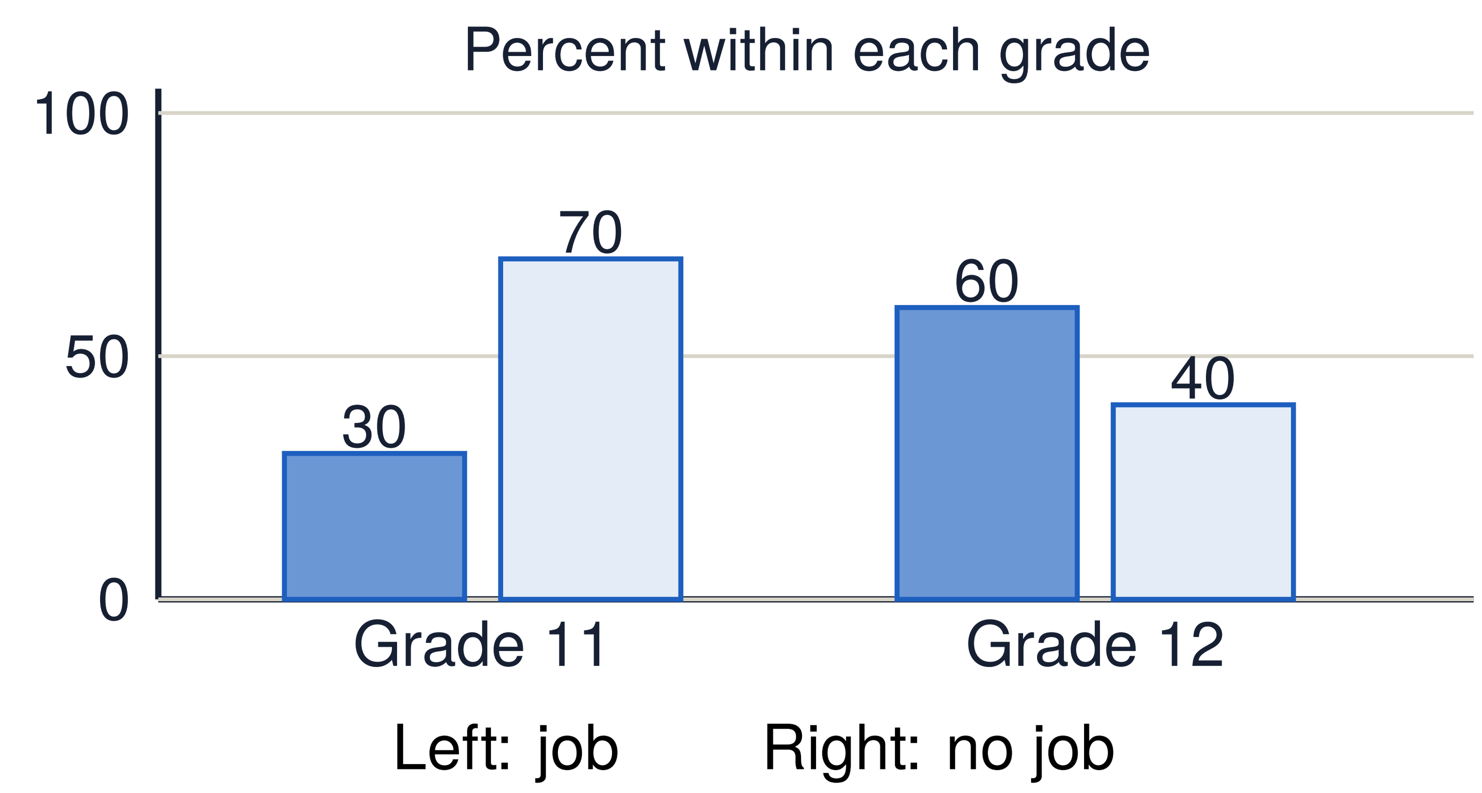
$$\text{marginal rel. freq.} = \frac{\text{row or column total}}{\text{grand total}}$$

$$\text{conditional rel. freq.} = \frac{\text{cell}}{\text{row or column total}}$$

SEGMENTED BARS: EACH GROUP IS 100%



SIDE-BY-SIDE BARS: COMPARE SHARES



ZONE B • WHAT THE WORDS MEAN

Explanatory variable: defines the groups we compare (grade level).
Response variable: the outcome compared (job status).

CONDITIONAL ON = WITHIN

Restrict the total to that row or column.
Each conditional distribution adds to 100%.

ZONE C • SAY IT IN WORDS

"There is an association because the conditional distribution of ____________ differs across ____________ groups: _____% versus _____%."

Name the response, the explanatory groups, and the compared category.

ZONE D • WATCH OUT

COMPARE CONDITIONAL DISTRIBUTIONS

Use shares, not counts, when group sizes differ.
State which group supplies the denominator.

see WS old u2_l1–3